Urban Air Quality Synthetic Dataset

Team: Pit Crew

Use Case: Urban Air Quality Dashboard

Target User: Government Operations / Urban Services Analyst

1. Objective

This project delivers a reusable analytics pipeline for urban air quality monitoring. The pipeline takes in station-level pollution readings and produces exploratory data analysis, visualizations, and predictive models for two operational tasks: forecasting next-period concentrations and predicting AQI exceedance events.

The pipeline is delivered with a synthetic dataset generator that produces realistic example data, allowing the analytics workflow to be developed, tested, and demonstrated end-to-end without ever requiring access to a client's real monitoring data. When a client is ready to use the pipeline on their own data, they replace the synthetic input with their CSV through a single configuration change. Everything downstream — distributions, diurnal patterns, weather correlations, regression and classification models — runs identically against client data.

The design supports the following downstream uses:

- Monitoring pollution levels across stations.
- Detecting hazardous air quality conditions through AQI exceedance modeling.
- Understanding environmental drivers (weather, traffic patterns, area characteristics).
- Supporting policy and operational decisions with quantitative evidence.

The synthetic data is engineered to mimic real-world characteristics so the pipeline can be calibrated against it: temporal dynamics with diurnal and weekly cycles, spatial dependencies via wind-driven transport, measurement imperfections including non-random missingness, and latent confounders representing traffic and industrial activity.

2. Deliverables

The project deliverable is a single Jupyter notebook divided into three operational layers:

2.1 Generator layer (cells 1–14)

A hierarchical spatiotemporal synthetic data generator that produces six relational tables — Cities, Stations, Pollutants, Weather, Readings, and Alerts — calibrated against EPA AQI standards and WHO air quality guidelines.

2.2 Data pipeline layer (cells 15–17)

A unified data-loading layer with two modes:

- Synthetic mode runs the generator and produces the demonstration dataset.
- Client mode loads a client-supplied CSV, normalizes its column names and pollutant identifiers, validates its schema, and produces a dataframe identical in shape to the synthetic output.

The mode switch is a single variable assignment in cell 17, the only cell a client edits to use their own data.

2.3 Analytics layer (cells 21–28)

A defensive EDA and modeling pipeline that produces the following outputs:

- Distribution analysis (histograms, log-scale boxplots per pollutant).
- Temporal analysis (diurnal patterns with confidence bands, time-series for sample stations).
- Spatial analysis (by area type, by city).
- Weather correlation analysis (heatmap of pollution vs. weather drivers).
- Regression model predicting concentration values (linear baseline + Random Forest).
- Classifier predicting AQI > 100 exceedance events (Random Forest with class balancing).

Each cell in this layer skips gracefully and prints an informative message when its required input columns are not present in the client's data, allowing the pipeline to run usefully on partial data without crashing.

3. Data Schema

3.1 Cities

Feature	Type	Description
city_id	int	Unique city identifier
population	int	Total population
urban_density	float	Population per km²
coastal	binary	Coastal indicator (1 = coastal)
avg_income	float	Proxy for infrastructure level

center_lat	float	Synthetic city-center latitude (used for station clustering)
center_lon	float	Synthetic city-center longitude

3.2 Stations

Feature	Type	Description
station_id	int	Unique station identifier
city_id	int	Foreign key to Cities
latitude	float	Station latitude (clustered around city_center_lat)
longitude	float	Station longitude (clustered around city_center_lon)
area_type	category	Residential, roadside, or industrial
maintenance_score	float	Sensor upkeep quality (0.6–1.0)
install_year	int	Installation year (used for sensor drift modeling)

3.3 Pollutants

Feature	Type	Description
pollutant_id	int	Unique pollutant identifier
name	category	PM2.5, PM10, NO2, O3, or CO
unit	string	Native measurement unit (μg/m³ or mg/m³)
safe_limit	float	Regulatory threshold (WHO/EPA-aligned)
hazard_weight	float	AQI contribution weight

3.4 Weather

Feature	Type	Description
city_id	int	Foreign key
datetime	timestamp	Hourly timestamp
temperature	float	Air temperature (°C) — diurnal + seasonal pattern

humidity	float	Relative humidity (0.6–0.95)
wind_speed	float	Wind speed (m/s) — lognormal-distributed
wind_direction	float	Wind direction (radians) — random walk on the circle
rainfall	float	Rainfall intensity (mm/hour) — zero-inflated

3.5 Readings (fact table)

Feature	Type	Description
reading_id	int	Unique reading identifier (primary key)
station_id	int	Foreign key
city_id	int	Foreign key
datetime	timestamp	Reading timestamp (hourly)
pollutant_id	int	Foreign key
concentration	float	True (latent) concentration value
reported_value	float	Observed (noisy) value; NaN when sensor offline
sensor_status	category	OK, drift, or offline

3.6 Alerts

Feature	Type	Description
alert_id	int	Unique alert identifier
station_id	int	Foreign key
datetime	timestamp	Alert trigger timestamp
pollutant_id	int	Worst pollutant at trigger time
alert_level	category	AQI band (Good through Hazardous)
aqi	float	Computed AQI value
delay_minutes	int	Reporting delay (Poisson-distributed, λ = 15)

4. Latent Variables

The following variables are generated to introduce realistic confounding and hidden structure but are not exposed in the final dataset. Their effects are observable only through their downstream influence on concentrations, missingness, and alerts.

Variable	Level	Purpose
traffic_intensity	station × hour	Drives traffic-related emissions; carries a two-bump diurnal pattern (rush hour) and a weekday/weekend modulation
industrial_activity	city × hour	Captures economic activity; weekend dampening
atmospheric_dispersion	city × hour	Weather-based dilution combining wind speed, rainfall, and humidity
sensor_quality	station	Affects measurement noise and probability of missingness via the maintenance_score

5. Structural Equations

The generator combines emissions modeling, atmospheric dispersion, wind-driven spatial transport, autoregressive temporal dynamics, and measurement-error modeling. Each component is documented below with the equation used in the implementation.

5.1 Emissions

Per-station emissions at time t aggregate traffic, industrial, and area-type contributions:

$$E_{i,t} = \alpha_1 \cdot \text{Traffic}_{i,t} + \alpha_2 \cdot \text{Industrial}_{c,t} + \alpha_3 \cdot \text{AreaType}_i$$

Pollutant-specific scaling produces five emission streams per station:

- PM2.5 and PM10 emissions are industrial-dominant: 0.5 × E_base + 0.5 × (3 × Industrial).
- NO2 and CO emissions are traffic-dominant: 0.7 × E_base + 0.3 × (5 × Traffic).
- O3 emissions follow a special photochemical equation (§5.6) and do not use this branch.

5.2 Atmospheric Dispersion

Dispersion is the inverse of concentration: high values dilute pollution, low values trap it. The implementation combines wind speed, rainfall, and humidity:

$$\text{Dispersion}_{c,t} = 0.5 \cdot \text{WindSpeed} + 0.3 \cdot \text{Rainfall} + 0.2 / \text{Humidity} + 0.5$$

A floor of 0.3 prevents pathological values when all three drivers are simultaneously low.

5.3 Spatial Transport

Pollutants travel from upwind stations to downwind stations along the prevailing wind direction. The contribution of station j to station i at time t is weighted by the alignment between the wind at j and the bearing from j to i, divided by the squared distance between them:

$$\text{Transport}_{i,t} = \Sigma_j \text{Alignment}_{i,j,t} \cdot C_{j,t-1} / (1 + d_{ij}^{\kappa})$$

$$\text{Alignment}_{i,j,t} = \max(0, \cos(\theta_{wind} - \theta_{j \to i}))$$

In the implementation the weight matrix is row-normalized so the transport term is a weighted average of neighbor concentrations rather than an unbounded sum. This stabilizes the AR(1) feedback loop in §5.5 across any choice of station density or transport coefficient.

5.4 Predicted Concentration

The predicted instantaneous concentration of pollutant p at station i and time t combines local emissions (scaled inversely by dispersion) with the spatial transport contribution:

$$\hat{C}_{i,t,p} = \gamma_p \cdot E_{i,t} / \text{Dispersion}_{c,t} + \delta_p \cdot \text{Transport}_{i,t}$$

γ_p is pollutant-specific: PM and NO_2 have different emission-to-concentration mappings reflecting their different formation pathways. δ_p is larger for fine particles than for reactive gases, reflecting that particulates have longer atmospheric residence times.

5.5 Temporal Autocorrelation

The actual concentration at time t is an AR(1) update combining the previous-step concentration with the predicted instantaneous concentration:

$$C_{i,t,p} = \varphi_p \cdot C_{i,t-1,p} + (1 - \varphi_p) \cdot \hat{C}_{i,t,p} + \varepsilon_p$$

φ_p is the persistence coefficient: higher for particulates (0.70–0.75), which have longer atmospheric residence, and lower for reactive gases (0.60–0.65). The innovation noise ε_p is pollutant-specific (Gaussian).

5.6 Ozone (Special Case)

Tropospheric ozone is not directly emitted; it forms via photochemical reactions involving solar radiation, nitrogen oxides, and volatile organic compounds. The model represents O_3 as increasing with temperature and decreasing with NO_2 (the NO_x-titration effect):

$$C^{O3}_{i,t} = \delta_1 \cdot \text{Temperature}_{c,t} - \delta_2 \cdot \text{NO2}_{i,t} + \varepsilon$$

The implementation computes NO_2 first within each timestep and uses that value when generating O_3 in the same step, capturing the within-hour NO-O_3 interaction.

5.7 Measurement Model

Reported (observed) values differ from true concentrations through additive sensor noise and slow calibration drift:

$$\text{Reported} = \text{Round}(C + \varepsilon\_\text{sensor} + \text{Drift}, 1)$$

Drift grows linearly with station age and elapsed time:

$$\text{Drift} = 0.02 \cdot (2024 - \text{install\_year}) \cdot (\text{days\_elapsed} / 30)$$

5.8 Missingness Model

Missingness is non-random: lower-quality sensors miss more readings, and rainfall events temporarily disrupt sensors:

$$P(\text{Missing}) = \sigma(-3 + 3 \cdot (1 - \text{SensorQuality}) + \text{WeatherShock})$$

WeatherShock is 1.5 when rainfall exceeds 10 mm/hour and zero otherwise. The intercept is calibrated to produce the target overall missing rate of 5–15%.

5.9 AQI Computation

AQI is computed by piecewise-linear interpolation between U.S. EPA breakpoints. For a concentration C falling in the range [C_low, C_high]:

$$\text{AQI\_p} = (\text{I\_high} - \text{I\_low})/(\text{C\_high} - \text{C\_low}) \cdot (C - \text{C\_low}) + \text{I\_low}$$

The overall station AQI at any time is the maximum across pollutants:

$$\text{AQI} = \text{max\_p}\ (\text{AQI\_p})$$

Full breakpoint tables for all five pollutants are documented in Section 6.

5.10 Alert Generation

An alert fires when the per-station AQI at any timestep exceeds 100 (the boundary between Moderate and Unhealthy for Sensitive Groups). Each alert records the worst pollutant at the trigger time and a Poisson-distributed reporting delay:

$$\text{Alert\_\{i,t\}} = \mathbb{1}(\text{max\_p AQI\_\{p,i,t\}} > 100)$$

$$\text{Delay\_minutes} \sim \text{Poisson}(\lambda = 15)$$

6. Parameter Calibration

The generator's parameters are calibrated against three target metrics: the mean concentration of each pollutant should fall in a realistic urban range, the overall AQI > 100 occurrence rate should fall between 10% and 25% of station-hours (matching real-world urban alert frequencies), and the missing-data rate should fall between 5% and 15%.

Calibration was achieved through analytical reasoning about the AR(1) steady state combined with empirical tuning. The values below produce a synthetic dataset that satisfies all three calibration targets on the default 5-city, 90-day, hourly configuration with a fixed random seed.

6.1 Reference parameter values

Group	Parameter	Description	Value
Emissions	α_1	Traffic emission weight	10.0
	α_2	Industrial emission weight	6.0
	α_3	Area-type weight (residential 1.0, roadside 1.5, industrial 1.8)	2.0
Concentration	γ_PM2.5	Emission-to-concentration scaling, PM2.5	0.3
	γ_PM10	Emission-to-concentration scaling, PM10	0.5
	γ_NO2	Emission-to-concentration scaling, NO2	0.5
	γ_CO	Emission-to-concentration scaling, CO (mg/m³)	0.025
Transport	δ_PM2.5	Transport coefficient, PM2.5	0.40
	δ_PM10	Transport coefficient, PM10	0.50
	δ_NO2	Transport coefficient, NO2	0.20
	δ_O3	Transport coefficient, O3	0.15
	δ_CO	Transport coefficient, CO	0.25
	κ	Distance decay exponent	2.0
Autocorrelation	φ_PM2.5 / φ_PM10	AR(1) memory for particulates	0.75 / 0.70
	φ_NO2	AR(1) memory for NO2	0.65
	φ_O3	AR(1) memory for O3	0.60
	φ_CO	AR(1) memory for CO	0.70

O3 chemistry	δ_1	Temperature coefficient (O3 formation)	1.2
	δ_2	NO2 coefficient (O3 titration)	0.4
Stochastic	spike probability	Heavy-tailed event probability per (i, t, p)	0.003
	Poisson λ	Mean alert reporting delay (minutes)	15

Background concentration floors (representing rural background and far-field transport) are 5.0, 12.0, 8.0, 25.0, and 0.4 in pollutant-native units for PM2.5, PM10, NO2, O3, and CO respectively. AR(1) innovation noise σ values are 3.0, 5.0, 4.0, 4.0, and 0.15 in the same order.

6.2 AQI breakpoint tables

The AQI is computed using U.S. EPA breakpoints (EPA, 2024). Tables for all five pollutants follow.

PM2.5 (μg/m³, 24-hr avg)

Concentration range	AQI range	Category
0.0 – 12.0	0 – 50	Good
12.1 – 35.4	51 – 100	Moderate
35.5 – 55.4	101 – 150	Unhealthy for Sensitive Groups
55.5 – 150.4	151 – 200	Unhealthy
150.5 – 250.4	201 – 300	Very Unhealthy
250.5 – 500.4	301 – 500	Hazardous

PM10 (μg/m³, 24-hr avg)

Concentration range	AQI range	Category
0 – 54	0 – 50	Good
55 – 154	51 – 100	Moderate
155 – 254	101 – 150	Unhealthy for Sensitive Groups
255 – 354	151 – 200	Unhealthy
355 – 424	201 – 300	Very Unhealthy

425 – 604	301 – 500	Hazardous

NO_2 (µg/m³, 1-hr avg)

Concentration range	AQI range	Category
0 – 53	0 – 50	Good
54 – 100	51 – 100	Moderate
101 – 360	101 – 150	Unhealthy for Sensitive Groups
361 – 649	151 – 200	Unhealthy
650 – 1249	201 – 300	Very Unhealthy
1250 – 2049	301 – 500	Hazardous

O_3 (µg/m³, 8-hr avg)

Concentration range	AQI range	Category
0 – 54	0 – 50	Good
55 – 70	51 – 100	Moderate
71 – 85	101 – 150	Unhealthy for Sensitive Groups
86 – 105	151 – 200	Unhealthy
106 – 200	201 – 300	Very Unhealthy

CO (mg/m³, 8-hr avg)

Concentration range	AQI range	Category
0.0 – 4.4	0 – 50	Good
4.5 – 9.4	51 – 100	Moderate
9.5 – 12.4	101 – 150	Unhealthy for Sensitive Groups
12.5 – 15.4	151 – 200	Unhealthy
15.5 – 30.4	201 – 300	Very Unhealthy

30.5 – 50.4	301 – 500	Hazardous

7. Validation Results

Validation was conducted on a fixed random seed (SEED = 42) producing a 90-day, 5-city, 63-station, 5-pollutant dataset of 680,400 readings. All calibration targets are met.

Metric	Observed	Target	Status
PM2.5 mean	29.26 µg/m³	25 – 60 µg/m³	Pass
PM10 mean	65.53 µg/m³	50 – 150 µg/m³	Pass
NO2 mean	47.64 µg/m³	30 – 80 µg/m³	Pass
O3 mean	46.71 µg/m³	30 – 60 µg/m³	Pass
CO mean	2.75 mg/m³	1 – 4 mg/m³	Pass
AQI > 100 rate	17.27 %	10 – 25 %	Pass
Missing rate	9.82 %	5 – 15 %	Pass

7.1 Modeling outcomes on synthetic data

The analytics layer produces the following outputs on the synthetic dataset:

Regression (concentration prediction):

```
Linear baseline MAE :   6.843   R²: 0.835
Random Forest  MAE :   5.100   R²: 0.902
RF improvement             : +25.5%
```

Classifier (AQI > 100 exceedance):

```
Train positives:  18,693 / 491,051  ( 3.81%)
Test  positives:   5,089 / 122,524  ( 4.15%)

          precision          recall  f1-score   support
 Below AQI 100            0.997      0.898      0.945      117435
          AQI > 100            0.283      0.929      0.434      5089

          accuracy                          0.899      122524
```

The classifier deliberately uses class_weight="balanced", producing high recall (0.93 — the model catches most alerts) at the cost of lower precision (0.28 — some false alarms). This trade-off is appropriate for a public-safety alerting context where missed alerts are more costly than false alarms.

On a per-row basis the positive class is 4% of test rows. This is consistent with the 17% per-station-hour AQI > 100 rate: each station-hour produces one row per pollutant (5 rows), but the alert fires on at most one of them (the worst pollutant at that hour), so the per-row positive rate is approximately 17% / 5 ≈ 3.4%.

8. Realism Features

The generated dataset exhibits the following structural properties expected of real-world urban air quality data:

- Hourly temporal dynamics with morning (≈8:00) and evening (≈18:00) rush-hour peaks in traffic-dominant pollutants (NO_2, CO).
- Weekly cycles, with weekend means roughly 30% lower than weekday means for traffic-driven pollutants.
- Seasonal-style temperature variation across the 90-day window plus a 24-hour diurnal temperature cycle.
- Spatial spillover: concentrations at downwind stations correlate more strongly with upwind sources than the reverse, captured by the wind-aligned transport term in §5.3.
- Heavy-tailed concentration spikes from rare pollution events (probability 0.003 per station-hour-pollutant).
- Sensor noise that is independent of true concentration.
- Slow calibration drift that grows with station age and elapsed time.
- Non-random missingness driven by sensor quality and rainfall events (§5.8).
- Reporting delays on alerts (Poisson-distributed, λ = 15 minutes).

9. Limitations

The dataset is intentionally simplified to remain tractable while preserving the structural features that matter for monitoring and analytics workflows. The principal limitations are as follows.

- Simplified atmospheric chemistry. The ozone model is a linear approximation of the temperature-NO_2 relationship rather than a full photochemical mechanism. Volatile organic compounds, aerosol partitioning, and secondary inorganic aerosol formation are not represented.
- Synthetic geography on a unitless coordinate grid. Station coordinates are sampled around city centers on an arbitrary 10 × 10 grid; distances are unitless,

and the transport distance decay is calibrated against this synthetic geometry rather than real-world plume dispersion data.

- Single-direction wind transport. Wind direction at each timestep is treated as a single per-city scalar applied uniformly across all stations in that city. Vertical mixing, boundary-layer stratification, sea-breeze circulations, and topographic channelling are not modeled.
- AR(1) temporal structure only. Pollutant time series follow a first-order autoregressive update with pollutant-specific persistence coefficients. This captures short-memory diurnal-scale autocorrelation but not moving-average components, longer-memory seasonal trends, or regime-switching.
- Limited pollutant interactions. Only the O_3-NO_2 coupling is represented; SO_2, ammonia, and primary biogenic emissions are absent. Cross-pollutant correlations arise primarily from shared driver variables (traffic, weather) rather than from chemical coupling.
- AQI computed on hourly readings. The pipeline applies EPA breakpoints (which are technically defined on 24-hour averages for PM2.5 and PM10, 8-hour averages for O_3 and CO, 1-hour averages for NO_2) directly to hourly readings. This produces conservative band assignments compared to a strict EPA-compliant pipeline. Users requiring strict regulatory comparability should aggregate to the appropriate window before AQI assignment.
- EPA breakpoints applied uniformly. WHO guidelines, EU AQI, and the Philippine DENR Air Quality Index use different thresholds. Studies requiring jurisdiction-specific AQI must remap concentrations against the corresponding regional breakpoints.
- Ninety-day default observation window. Too short for robust seasonal pattern recovery; the synthetic dataset implicitly represents a single quarter. Users requiring annual cycles or holiday effects should extend n_days and recalibrate.
- Exogenous emission drivers. Traffic intensity and industrial activity are sampled from fixed lognormal distributions modulated by deterministic diurnal and weekly patterns. The model does not represent endogenous responses such as policy interventions, fuel-mix transitions, mode-shift effects, or behavioral adaptation to alerts.
- Linear sensor drift. Calibration drift is modeled as a smooth linear function of station age and elapsed time. Real sensors often exhibit non-linear, episodic drift events such as power cycles, recalibration interventions, and component failures.
- Synthetic data is more predictable than real data. Random Forest models trained on the synthetic data achieve high R^2 because the feature set fully observes the generative process inputs. Real-world pollution depends on unobserved confounders, so models trained on real data will typically underperform the synthetic benchmark — the synthetic R^2 should be interpreted as an upper bound of what the same modeling pipeline can achieve.

10. Client Handoff Guide

A client uses this pipeline by editing a single cell. This section documents the input contract and the operational workflow.

10.1 Required input format

The client provides a CSV with at least the four required columns below. Column names must match exactly, or the client passes a column_map dict to remap them.

Column	Type	Description
station_id	str or int	Unique identifier per monitoring station
datetime	parseable date	Reading timestamp (hourly recommended)
pollutant	str	Pollutant name (PM2.5, NO2, etc. — see §10.2)
value	float	Concentration in source's native unit

Optional columns extend what the analytics layer can do. Each is independent — the pipeline uses what's provided and skips cells whose required columns are absent:

Column	Type	Used by
city_id	str/int	By-city plot, regression features
area_type	str	By-area boxplot, regression features
temperature	float	Weather correlation, regression features
wind_speed	float	Weather correlation, regression features
humidity	float	Weather correlation, regression features
rainfall	float	Weather correlation, regression features
maintenance_score	float	Regression features

10.2 Pollutant naming

The pipeline normalizes common pollutant-name variants automatically. Variants such as 'pm25', 'PM_2_5', 'PM 2.5', and 'particulate_matter_2_5' all map to the canonical 'PM2.5' used by the AQI lookup. Supported canonical names are PM2.5, PM10, NO2, O3, and CO. Pollutants outside this set pass through unchanged with a one-time warning that AQI will not be computed for them.

10.3 Operational workflow

To run the pipeline on client data:

- Place the client CSV in the data/ folder.
- Open cell 17 in the notebook. Change MODE = "synthetic" to MODE = "client".
- In the same cell, set the path argument of load_client_data to point to the client CSV. If the client's column names differ from the standard four required names, supply a column_map dict, e.g., column_map={"value": "concentration_ugm3"}.
- Run cells 17 onward. Cell 17 produces a schema report so the client immediately sees how their data was parsed.
- All EDA cells (21–25) and modeling cells (27–28) run automatically, skipping cells whose required columns are absent.

10.4 Pipeline outputs

The notebook produces the following client-facing outputs when run on either synthetic or client data:

- A schema report listing rows, time range, granularity, station count, city count, pollutant set, missing rate, and which optional columns are present.
- A distribution histogram and per-pollutant log-scale boxplot.
- A diurnal-pattern lineplot with 95% confidence bands.
- A by-area-type boxplot (if area_type is provided).
- A by-city barplot (if more than one city is present).
- A weather correlation heatmap (if any weather columns are provided).
- A two-week hourly time series for a representative station and pollutant.
- Regression predictions of concentration with linear and Random Forest baselines, MAE, R^2, feature importance, and a residual plot.
- AQI > 100 classifier with full precision/recall/F1 report and confusion matrix.

10.5 Pipeline limitations for client data

- The pipeline assumes hourly granularity. Daily-resolution data will run but the diurnal plot collapses to a single point.
- The classifier requires at least 10 AQI > 100 events in both train and test sets. Lightly polluted areas will trigger the no-positives skip path.
- AQI computations use EPA breakpoints. Clients in jurisdictions with different thresholds (WHO, EU, DENR) will need to remap.
- Modeling features one-hot-encode pollutant identity, area type, and city. Clients with very large numbers of any of these (e.g., hundreds of stations identified individually) may want to bucket or aggregate before loading.

11. References

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12. Validation Plan

Realism claims in §8 are operationalized as a battery of automated checks. Failure of any check indicates the calibration constants in §6 must be retuned before the dataset is released for downstream use.

12.1 Distributional checks

Check	Acceptance band	Justification
PM2.5 24-hour mean	25.0 – 60.0 µg/m³	Aligns with WHO 2021 interim target 1 (35 µg/m³) and observed urban means
AQI > 100 occurrence rate	10 – 25 % of station-hours	Matches typical major-city alert frequencies
Reading missingness	5 – 15 % of all readings	§6 calibration target

Per-pollutant 95th percentile	Within 2× of AQI 300 breakpoint	Heavy tails should reach 'Very Unhealthy' but not exceed sensor saturation
Per-pollutant minimum	≥ 0	Concentrations are non-negative

12.2 Temporal checks

Lag-1 autocorrelation for PM2.5 (averaged across stations) should fall in the range [0.55, 0.85], consistent with the AR(1) coefficient of 0.75. Hourly aggregates of NO2 and CO concentrations should show twin peaks at approximately 8:00 and 18:00. Weekday-versus-weekend means for traffic-driven pollutants (NO2, CO) should differ by at least 20 percent.

12.3 Spatial checks

Within-city station-pair correlations of hourly readings should exceed between-city pair correlations by a clear margin. For station pairs where one lies downwind of the other under the prevailing wind direction, the lag-1 cross-correlation from upwind to downwind should exceed the reverse, providing direct evidence that the wind-direction-aware transport term is active.

12.4 Missing-data integrity checks

The Pearson correlation between station-level maintenance_score and station-level missing rate must be significantly negative ($p < 0.05$): stations with lower maintenance scores must miss more readings. The correlation between heavy-rainfall events (rainfall > 10 mm/hour) and missingness must be positive.

12.5 Reporting and remediation

The implementation includes a validate() function that runs all distributional checks and prints results alongside their acceptance bands. When any check fails, the report identifies which calibration constants in §6 are most likely responsible. Generation runs whose datasets fail validation are not released; calibration constants are tuned, and the pipeline is re-run from a fixed random seed to preserve reproducibility.